A direct larger-noise comparison from crossing cancellation

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The crossing identity gives a nonnegative loss, but its sign alone does not pay for the finite binomial and normal corrections. Here is a quantitative version. A characteristic-function estimate measures the loss for one fixed count; a coupling transfers it to the selected count. The inward displacement makes this transfer inexpensive. Only a fixed window of counts then needs a lower binomial-mass estimate. The rest is bounded by a summable fourth moment.

Theorem 1. *Let $n \ge 6000$ be an integer, $p \in [2/5, 53/100]$, $q = 1-p$, and $Y = B + U$, with B Bernoulli of parameter p and $\mathbb{E}(U \mid B) = 0$. Assume*

$$-\tfrac{3}{5} \le U_0 \le a_0, \quad -a_1 \le U_1 \le \tfrac{3}{5}, \quad a_0, a_1 \ge 0, \quad d = a_0 + a_1 \le \min\{9/125, 3/n\}.$$

Put $v = pq$, $s = \mathbb{E}U^2$, $\lambda = ns$, $X = (Y-p)/\sqrt{v+s}$, $\rho = \mathbb{E}|X|^3$ and $\rho_B = (1-2v)/\sqrt{v}$. Suppose $3/40 \le \lambda \le 9/25$ and $|H| \le 46/25$. Write

$$t = np + H\sqrt{v+s}, \quad k = \lfloor t \rfloor, \quad u = t - k, \quad \delta = k - np.$$

For every $C \ge 0$, with F_B the $\mathrm{Bin}(n,p)$ *CDF,*

$$\begin{aligned}\Delta_C = \sqrt{n}\{\mathbb{P}(X_1 + \cdots + X_n \le H) - \Phi(H/\sqrt{n})\} - C\rho \\ - \sqrt{n}\{F_B(k) - \Phi(\delta/\sqrt{nv})\} + C\rho_B < -\lambda/200.\end{aligned}$$

This is a local comparison for all laws in the stated strip. It is not an unconditional Berry–Esseen cutoff theorem. Its application to a putative maximizing witness retains the actual-gap acquisition premises. The finite base and the intermediate indices remain separate obligations.

1 The exact crossing formula

Use the independent conditional pairs and increasing mesh from `simplification-v13/2026-10-28/crossing.tex`:

$$T_j = j + W_j = j + \sum_{i \le j} U_{1i} + \sum_{i > j} U_{0i}, \qquad T_K \le t < T_{K+1}.$$

Here $T_0 \le t < T_n$: indeed $v \ge 6/25$, $\sqrt{v+s} < .501$, and $|t - np| < .94$. These estimates also give $|\delta| < 2$. Define

$$h(x) = \mathbb{E}(x - U_0)_+ - \mathbb{E}(x - 1 - U_1)_+ - \mathbb{P}(1 + U_1 \le x), \quad R_i = t - (i-1) - \sum_{l<i} U_{1l} - \sum_{l>i} U_{0l}.$$

The preceding finite telescoping proof gives, with $g_i = \mathbb{E}h(R_i)$,

$$g := \sum_{i=1}^{n} g_i = t - \mathbb{E}K \ge \mathbb{E}\min\{t - T_K, 41/125\}, \qquad g_i \ge 0. \tag{1}$$

In fact $0 \le h \le 1$, and h vanishes outside $[-3/5, 8/5]$. Below $1 - a_1$, it is the increasing function $\mathbb{E}(x - U_0)_+$, whose limit there is $1 - a_1 \le 1$; above that point it equals $\mathbb{P}(1 + U_1 > x) - \mathbb{E}(1 + U_1 - x)_+ \le 1$.

With $\pi_j = \mathbb{P}(B_n = j)$, put

$$B_j(t) = \mathbb{E}(t - T_j)_+ - (t - j)_+, \qquad \mathcal{C}(t) = \sum_{j=1}^{n-1} (\pi_{j+1} - \pi_j) B_j(t).$$

The exact weighted formula is

$$\Delta_C = -\sqrt{n} \sum_i \pi_i g_i + \sqrt{n}\mathcal{C}(t) + \mathcal{P} + C(\rho_B - \rho), \tag{2}$$

where

$$\mathcal{P} = \sqrt{n} \left[u\pi_{k+1} - \Phi\left(\frac{\delta + u}{\sqrt{n(v+s)}} \right) + \Phi\left(\frac{\delta}{\sqrt{nv}} \right) \right].$$

All lower crossing events are closed and upper crossing events strict. The identities apply to atoms without a continuity assumption.

2 Transport from the selected count to a fixed count

Lemma 2. *On this one common probability space,*

$$\mathbb{E}|W_K - W_k| < 13d.$$

Proof. Let $D_i = U_{1i} - U_{0i}$, and $V_i = \sum_{l<i} U_{1l} + \sum_{l>i} U_{0l}$. Centering and the inward caps give $\mathbb{E}|D_i| \le 2d$; V_i is independent of this pair and has variance at most $n \max(v_0, v_1) \le (5/2)\lambda \le 9/10$, where $v_j = \mathbb{E}U_j^2$.

Expand $W_K - W_k$ into $D_i \mathbf{1}_{\{K \ge i\}}$ for $i > k$ and $-D_i \mathbf{1}_{\{K < i\}}$ for $i \le k$ and use the triangle inequality. For $i = k + r$, $r \ge 1$, the event $K \ge i$ is $T_i \le t$, which implies $V_i \le u - r + a_1$. For $r = 1$ use probability at most one; for $r \ge 2$ use Chebyshev with $r - 1 - d > 0$. For $i = k - r$, $r \ge 0$, the event $K < i$ is $T_i > t$, which implies $V_i > u + r - 3/5$. Use probability at most one for $r = 0, 1$ and Chebyshev with $r - 3/5 > 0$ thereafter. The endpoint replacements make these events depend only on V_i, so their probabilities multiply $\mathbb{E}|D_i|$; no stopped-index independence is asserted.

For a decreasing positive function f, $\sum_{r=m}^{\infty} f(r) \le f(m) + \int_m^{\infty} f(x)\,dx$. Consequently

$$\mathbb{E}|W_K - W_k| \le 2d \left[3 + \frac{9}{10} \left\{ \frac{60}{49} + \left(\frac{125}{116} \right)^2 + \frac{125}{116} \right\} \right] < 13d.$$

This bound includes both full tails and is uniform in n and the law. □

3 A quantitative unweighted loss

Lemma 3. *One has $g \ge \lambda/5 - 13d$.*

Proof. For $|x| \le 84/125$, the elementary bound

$$1 - \cos(2\pi x) \ge 3x^2 \tag{3}$$

holds. Here are rational checks avoiding any numerical trigonometry. On $0 \le x \le 2/3$, concavity gives $\sin(\pi x) \ge (3\sqrt{3}/4)x > (51/40)x$. On $2/3 \le x \le 84/125$, the derivative bound $|\sin'| \le 1$

gives $\sin(\pi x) \ge 17/20 - 44/2625$. Squaring these positive lower bounds proves (3), using $\sqrt{3} > 17/10$ and $\pi < 22/7$.

If Z is either conditional noise and Z' an independent copy, then $|Z - Z'| \le 3/5 + d \le 84/125$. Thus

$$1 - |\mathbb{E}e^{2\pi i Z}|^2 = \mathbb{E}\{1 - \cos(2\pi(Z - Z'))\} \ge 6 \operatorname{Var} Z.$$

For $x \ge 0$ with $2x \le 1$, $(1 - 2x)(1 + x)^2 \le 1$. It follows that $|\mathbb{E}e^{2\pi i Z}| \le (1 + 3 \operatorname{Var} Z)^{-1}$. Independence and multiplication give

$$|\mathbb{E}e^{2\pi i W_k}| \le \frac{1}{1 + 3t_k}, \qquad t_k = \operatorname{Var} W_k \ge (1 - 5/n)\lambda \ge \frac{37}{42}\lambda.$$

The variance estimate follows directly by writing $t_k = (nq - \delta)v_0 + (np + \delta)v_1$ and using $|\delta| < 2$, $p, q \ge 2/5$.

For every $r \ge 0$, $\sin^2(\pi r) \le \min\{\pi r, 1\}$, hence $\min\{r, 41/125\} \ge (3/10)\sin^2(\pi r)$. Apply (1) with $r = t - T_K$. Since K is an integer,

$$\begin{aligned} g &\ge \frac{3}{20}\{1 - |\mathbb{E}e^{2\pi i W_K}|\} \\ &\ge \frac{3}{20}\left(1 - \frac{1}{1 + 3t_k}\right) - \frac{3\pi}{10}\mathbb{E}|W_K - W_k| \\ &\ge \frac{\lambda}{5} - 13d. \end{aligned}$$

In the last step use $3\pi/10 < 1$, Lemma 2, and $t_k \ge 37\lambda/42$, $\lambda \le 9/25$. The selected-mean bias is already included in (1); it has not been set to zero or paid for a second time. □

4 Restoring the binomial weights

For V_i above, independence and centering imply

$$\mathbb{E}V_i^4 \le (3/5)^2 \operatorname{Var} V_i + 3(\operatorname{Var} V_i)^2 \le \frac{153}{20}\lambda.$$

If $i = k + j$, $j \ge 11$, then $h(R_i) \ne 0$ requires $V_i \le u + 1 - j + 3/5 < 13/5 - j$. If $i = k - j$, $j \ge 11$, it requires $V_i \ge u + j - 3/5 \ge j - 3/5$. Since $0 \le h \le 1$, Markov's inequality and the decreasing-tail integral give

$$\begin{aligned} \sum_{|i-k|>10} g_i &\le \frac{153\lambda}{20} \sum_{j \ge 11} \{(j - 13/5)^{-4} + (j - 3/5)^{-4}\} \\ &\le \frac{153\lambda}{20}\left[(5/42)^4 + \frac{1}{3}(5/42)^3 + (5/52)^4 + \frac{1}{3}(5/52)^3\right] < \frac{\lambda}{100}. \end{aligned}$$

We record explicitly the finite factorial estimate used below:

$$\log(m!) = (m + \tfrac{1}{2})\log m - m + \tfrac{1}{2}\log(2\pi) + r_m, \qquad 0 < r_m < 1/(12m), \quad m \ge 1.$$

Its elementary integral-remainder proof is retained in `manuscript/arxiv-v1/appendices/count-comparison.tex`, the paragraph beginning "For completeness, the finite Stirling formula used here is". We use neither a limiting local CLT nor an unspecified asymptotic remainder.

If $|i-k| \le 10$, then $|i-np| < 12$. For $z = i/n$, every point between z and p has variance product at least $23/100$. The relative entropy $D(z\|p)$ therefore satisfies $nD(z\|p) \le (i-np)^2/(2n(23/100))$. The factorial estimate gives

$$\sqrt{n}\pi_i \ge \frac{\phi(0)}{\sqrt{z(1-z)}} \exp\left\{-\frac{144/2 + 1/12}{6000(23/100)}\right\} > \frac{199}{250}\left(1 - \frac{53}{1000}\right) > \frac{3}{4}.$$

Here $z \in [.398, .532]$ and $\phi(0) > 199/500$; all bounds are rational apart from the displayed standard constants, whose enclosures are checked. Combining this with Lemma 3 and $d \le 3/n$ gives

$$\sqrt{n}\sum_i \pi_i g_i > \frac{3}{4}\left(\frac{19\lambda}{100} - \frac{39}{n}\right) = \frac{57\lambda}{400} - \frac{117}{4n}. \tag{4}$$

5 The finite corrections

The signed-curvature lemma in `simplification-v13/2026-10-28/counts-tools.tex` gives

$$\sqrt{n}\mathcal{C}(t) \le \frac{\lambda}{4v\sqrt{n}} + \frac{c\sqrt{\lambda}}{2v\sqrt{n+1}}, \qquad c = (np - q - t)_+ < .46.$$

Since $v \ge 6/25$, this is at most

$$\frac{25\lambda}{24\sqrt{n}} + \frac{23\sqrt{\lambda}}{24\sqrt{n+1}}. \tag{5}$$

We next show $\mathcal{P} < 5/n$. Put $\xi = t - np$, so $|\xi| < .94$ and $\delta = \xi - u$. At $z = (k+1)/n$, the segment to p lies in $[.399, .531]$. On this segment

$$|(\log(z(1-z)))'| < 1, \qquad |D'''(z\|p)| < 4.$$

The factorial estimate and Taylor's theorem consequently give

$$\sqrt{n}\pi_{k+1} \le \frac{\phi(0)}{\sqrt{v}} \exp\left\{-\frac{(\delta+1)^2}{2nv} + e\right\}, \quad ne \le 1 + \frac{1}{12} + \frac{16}{3n} < \frac{11}{10}. \tag{6}$$

For $0 \le x \le u$, divide this by $f(x) = \phi((\delta+x)/\sqrt{n(v+s)})/\sqrt{v+s}$. Its logarithmic upper bound is

$$\frac{1}{n}\left[\frac{37}{20} + \frac{(1-x)(2u + 2(47/50) - x - 1)}{2v}\right] =: \frac{L(x,u)}{n}.$$

The first constant is $11/10 + \lambda/(2v) \le 11/10 + 3/4$. The bracket is nonnegative and less than eight. Also $f(x) < 41/50$. Using $e^z - 1 \le z/(1-z)$ for $0 \le z < 1$, integration gives

$$\sqrt{n}\pi_{k+1}u - \int_0^u f(x)\,dx \le \frac{41}{50n}\frac{750}{749}\left\{\frac{37}{20} + \frac{191}{72}\right\} < \frac{15}{4n}.$$

For the integral of the quadratic part, its numerator is $u[(2-u)(47/50) + 2u - (2/3)u^2 - 1]$, which increases on $[0,1]$ and is at most $47/50 + 1/3 = 191/150$. Finally changing the normal variance at the anchor contributes adversely only for $\delta > 0$, and then contributes less than

$$\frac{\phi(0)(47/50)(9/25)}{2(6/25)^{3/2}n} < \frac{1}{n}.$$

This proves $\mathcal{P} < 5/n$ with its full $1/n$ dependence.

The marginal cubic correction is favorable. Let $v_j = \mathbb{E}U_j^2$ and $\mu_j = \mathbb{E}U_j^3$. Since the two supports have their indicated signs relative to p, the raw absolute third moment is

$$v(1-2v) + 3v(v_0 + v_1) - q\mu_0 + p\mu_1.$$

The one-sided bounds $\mu_0 \le a_0 v_0$ and $\mu_1 \ge -a_1 v_1$ give an increase at least $(3\min(p,q) - d)s \ge (141/125)s$. The inequality $(1+x)^{3/2} \le 1 + 3x/2 + 3x^2/8$ for $x \ge 0$ then yields

$$(v+s)^{3/2}(\rho - \rho_B) \ge s\{87/250 - (13/16)s\} > 0.$$

Thus $C(\rho_B - \rho) \le 0$ for every $C \ge 0$.

Substitute these bounds into (2):

$$\Delta_C < -\frac{57\lambda}{400} + \frac{137}{4n} + \frac{25\lambda}{24\sqrt{n}} + \frac{23\sqrt{\lambda}}{24\sqrt{n+1}} < -\frac{\lambda}{200}.$$

For the final inequality divide by λ, use $n \ge 6000$, $\lambda \ge 3/40$, $\sqrt{n}, \sqrt{n+1} > 77$ and $\sqrt{\lambda} > 27/100$. The resulting strict rational inequality is included in the companion check. This proves Theorem 1.

6 Why the acquired displacement is small enough

For an actual central-gap witness, use the notation and the expansion before AM–GM in `simplification-v13/2026-10-28/geometry-noise.tex`. Here D denotes that geometric cost, not the conditional variable above. On the right strip every noise coefficient in that lower budget is positive; $Q(p) \ge 0$ and the reciprocal-cubic terms are nonnegative. Hence

$$nD \ge \frac{6nv\, d(1-d)}{(v+s)^{3/2}}.$$

If in addition the retained coarse budget gives $nD \le 129/4$, then $n \ge 6000$, $\lambda \le 9/25$ and $d \le 9/125$ imply

$$nd \le \frac{(129/4)(v+s)^{3/2}}{6v(1-d)} < \frac{(129/4)(251/500)}{6(116/125)} < 3.$$

The actual prior bound 32.20665378 is smaller than 129/4. This is the explicit acquisition join for the new theorem; it does not replace acquisition, reflection, attainment or the finite base.

For the sharp coefficient, the retained v12 comparison covers smaller positive noise. Thus within the acquired right strip the new theorem removes the entire remaining noise range when $n \ge 6000$. The v13 geometric index restriction already excludes acquired $p \le .4$ at such indices. The range $42 \le n < 6000$ and the independent finite base remain open. No unconditional numerical cutoff for the conjecture is asserted.